Abdul Quadir

Department of Mathematics
Indian Institute of Technology Indore, India

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Education

Research Associate Aug. 2026 – Present
Indian Institute of Technology Indore Indore, India
– **Advisor**: Prof. M. Tanveer

Doctor of Philosophy (Ph.D.) in Machine Learning Jul. 2022 – Jul. 2026
Indian Institute of Technology Indore Indore, India
– **Supervisor**: Prof. M. Tanveer, Department of Mathematics
– **Research Interests**: Vision Transformer, Graph Neural Networks (GNN), Randomized Neural Networks, Statistical Machine Learning, Granular Computing, Multi-View Learning, Trustworthy AI

Master of Science (M.Sc.) in Mathematical Science Jul. 2020 – Jun. 2022
Indian Institute of Technology Indore Indore, India

Bachelor of Science (B.Sc.) in Mathematical Science Jul. 2017 – Jun. 2020
Doon University Dehradun, India

Journal Publications

- **A. Quadir**, M. Sajid, and M. Tanveer. Granular ball twin support vector machine. *IEEE Transactions on Neural Networks and Learning Systems*, vol. 36, p. 12444-12453, 2025. doi:10.1109/TNNLS.2024.3476391.
- **A. Quadir** and M. Tanveer. Granular ball twin support vector machine with pinball loss function. *IEEE Transactions on Computational Social Systems*, vol. 12, p. 3891-3900, 2024. doi:10.1109/TCSS.2024.3411395.
- **A. Quadir** and M. Tanveer. Hypergraph neural network with state space models for node classification. *Engineering Applications of Artificial Intelligence*, Elsevier, vol. 163, p. 112922, 2026. 10.1016/j.engappai.2025.112922.
- **A. Quadir**, M. Sajid, M. Tanveer and P. N. Suganthan. Enhanced feature based granular ball twin support vector machine. *IEEE Transactions on Emerging Topics in Computational Intelligence*. arXiv preprint: https://arxiv.org/abs/2410.05786.
- M. Tanveer, M. Sajid, M. Akhtar, **A. Quadir**, T. Goel, A. Aimen, S. Mitra, Y-D Zhang, C. T. Lin, and J. Del Ser. Fuzzy deep learning for the diagnosis of Alzheimer's disease: Approaches and challenges. *IEEE Transactions on Fuzzy Systems*, vol. 32, p. 5477-5492, 2024. doi:10.1109/TFUZZ.2024.3409412.
- **A. Quadir** and M. Tanveer. TRKM: Twin Restricted Kernel Machines for Classification and Regression. *Neural Networks*, Elsevier, vol. 197, p. 108449, 2026. 10.1016/j.neunet.2025.108449.
- **A. Quadir**, M. Akhtar, and M. Tanveer. Enhancing multiview synergy: Robust learning by exploiting the wave loss function with consensus and complementarity principles. *Neural Networks*, Elsevier, vol. 188, p. 107433, 2025. doi:10.1016/j.neunet.2025.107433.
- **A. Quadir** and M. Tanveer. Multiview learning with twin parametric margin SVM. *Neural Networks*, vol. 180, p. 106598, 2024. doi:10.1016/j.neunet.2024.106598.

- **A. Quadir** and M. Tanveer. GARFLN: Geodesic Adaptive Riemannian Functional Link Network. *Pattern Recognition*, Elsevier, vol. 180, p. 114163, 2026. doi:10.1016/j.patcog.2026.114163.
- M. Sajid, **A. Quadir**, M. Tanveer, and Alzheimer's Disease Neuroimaging Initiative. GB-RVFL: Fusion of randomized neural network and granular ball computing. *Pattern Recognition*, Elsevier, vol. 159, p. 111142, 2025. doi:10.1016/j.patcog.2024.111142.
- M. Akhtar, **A. Quadir**, M. Tanveer, and Mohd. Arshad. Dual-Center RAPID-LSSVM: Radius-Adaptive, Probability and Imbalance Driven Weighting for Alzheimer's Diagnosis. *Neural Networks*, Elsevier, vol. 202, p. 108956, 2026. doi:10.1016/j.neunet.2026.108956.
- M. Tanveer, A. Mishra, **A. Quadir**, M. Sajid. EDA-OCBLS: An Error-Distribution Aware One-Class Broad Learning System for Anomaly Detection. *Neural Networks*, Elsevier, vol. 203, p. 109177, 2026. doi:10.1016/j.neunet.2026.109177.
- M. Tanveer, M. Pathak, M. Sajid, **A. Quadir**. Fuzzy-Driven Broad Learning System with Class Probability and Density Awareness for Multi-view Data. *Neural Networks*, Elsevier, vol. 201, p. 108914, 2026. doi:10.1016/j.neunet.2026.108914.
- M. Tanveer, A. Mishra, M. Sajid and **A. Quadir**. BLS-CIL: Class Imbalance Broad Learning System via Dual Weighting and Layer Trimming. *Pattern Recognition*, Elsevier, vol. 174, p. 112940, 2026. doi:10.1016/j.patcog.2025.112940.
- A. Akhtar, A. Kumari, M. Sajid, **A. Quadir** and M. Arshad, P. N. Suganthan and M. Tanveer. Towards robust and inversion-free randomized neural networks: The XG-RVFL framework. *Pattern Recognition*, Elsevier, vol. 172, p. 112711, 2026. doi:10.1016/j.patcog.2025.112711.
- M. Tanveer, R. K. Sharma, **A. Quadir** and M. Sajid. Enhancing robustness and efficiency of least square twin SVM via granular computing. *Pattern Recognition*, Elsevier, vol. 170, p. 112021, 2026. doi:10.1016/j.patcog.2025.112021.
- **A. Quadir**, M. Ganaie, and M. Tanveer. Intuitionistic fuzzy generalized eigenvalue proximal support vector machine. *Neurocomputing*, vol. 608, p. 128258, 2024. doi:10.1016/j.neucom.2024.128258.
- M. Tanveer, R. K. Sharma, M. Sajid and **A. Quadir**. GRVFL-MV: Graph random vector functional link based on multi-view learning. *Information Sciences*, Elsevier, vol. 704, p. 121947, 2025. doi:10.1016/j.ins.2025.121947.

Journal Manuscripts (Under Review / Revision)

- **A. Quadir**, and M. Tanveer. GraphFractalNet: A Fractal-Inspired Sparse Transformer for Ultra-Scalable Graph Representation Learning. *NeurIPS 2026*.
- **A. Quadir**, and M. Tanveer. ASTRIDE: Adaptive Semantic Token Routing with Hierarchical Context Distillation for Efficient Vision Transformers. *NeurIPS 2026*.
- **A. Quadir**, and M. Tanveer. SpectFormer: Joint Spatial-Spectral Cross-Covariance Attention for Alias-Free and Scale-Consistent Vision Transformers. *NeurIPS 2026*.
- **A. Quadir**, and M. Tanveer. AdaptiVision: A Flexible and Efficient Vision Transformer for Adaptive Token Pruning. *IEEE Transactions on Pattern Analysis and Machine Intelligence*. Revision Submitted.
- **A. Quadir**, and M. Tanveer. ALC-ViT: Adaptive Layer-Wise Convolutional Vision Transformer. *IEEE Transactions on Pattern Analysis and Machine Intelligence*.
- **A. Quadir**, and M. Tanveer. HBxLSTM-ASG: Hierarchical Bidirectional xLSTM with Adaptive Spatial Gating as a Generic Vision Backbone. *Transactions on Machine Learning Research (TMLR)*.
- **A. Quadir**, and M. Tanveer. GM-Mamba: Geometric Manifold Mamba for Visual Recognition. *IEEE Transactions on Neural Networks and Learning Systems*.

- **A. Quadir**, and M. Tanveer. Ensemble Broad Learning System with Graph Convolutional Kernel Machine. *IEEE Transactions on Neural Networks and Learning Systems*. Under Minor Revision.
- **A. Quadir** and M. Tanveer. HAMFNet: Hierarchical Attention-Guided Multi-Scale Feature Fusion for Medical Image Segmentation. *IEEE Transactions on Neural Networks and Learning Systems*.
- **A. Quadir** and M. Tanveer. GBTSVR: Twin Support Vector Regression via Granular Ball Computing. *IEEE Transactions on Neural Networks and Learning Systems*.
- **A. Quadir** and M. Tanveer. Graph Convolutional Restricted Kernel Machine for Classification. *IEEE Transactions on Cybernetics*.
- **A. Quadir**, M. Sajid and M. Tanveer. Multiview Random Vector Functional Link Network for Predicting DNA-Binding Protein. *IEEE Transactions on Consumer Electronics*. arXiv preprint: https://arxiv.org/abs/2409.02588.
- **A. Quadir** and M. Tanveer. Supervised Granular Ball Graph Coarsening with Random Vector Functional Link Network. *Pattern Recognition*.

Conference Publications

- M. Tanveer, Ayush Singh Rana, Sanskriti Jain, Arnav Kumar, Aryaman Tiwari, Abdur Rahaman, **A. Quadir**, M. Sajid. GraM-Diff: A Unified Graph–Mamba Diffusion Framework for EEG-Based Alzheimer's Disease Data Generation and Diagnosis. *MICCAI 2026 Workshop MSB EMERGE*.
- **A. Quadir**, A. Rahaman, M. Akhtar and M. Tanveer. Robust Dual-Model Collaborative Random Vector Functional Link Network. *IEEE World Congress on Computational Intelligence (WCCI)*, Netherlands, 2026.
- **A. Quadir**, A. Rahaman, P. N. Suganthan and M. Tanveer. GBFRVFL: Granular-Ball Computing-Based Fuzzy Random Vector Functional Link Network. *IEEE World Congress on Computational Intelligence (WCCI)*, Netherlands, 2026.
- A. Rahaman, **A. Quadir**, M. Sajid, M. Akhtar, and M. Tanveer. ECA-BLS: An Efficient Complex-Augmented Broad Learning System. *IEEE World Congress on Computational Intelligence (WCCI)*, Netherlands, 2026.
- A. Rahaman, **A. Quadir**, and M. Tanveer. RoBell-RVFL: A Robust Generalized Bell Random Vector Functional Link Network. *IEEE World Congress on Computational Intelligence (WCCI)*, Netherlands, 2026.
- M. Akhtar, A. Varshney, **A. Quadir**, A. Rahaman, M. Tanveer, and Mohd. Arshad. Robust Broad Learning System with Wave Loss for Classification under Data Uncertainty. *IEEE World Congress on Computational Intelligence (WCCI)*, Netherlands, 2026.
- **A. Quadir**, M. Sajid, M. Akhtar and M. Tanveer. Twin Restricted Kernel Machines for Multiview Classification. *International Joint Conference on Neural Networks (IJCNN)*, Rome, Italy, 2025. doi:10.1109/IJCNN64981.2025.11228247.
- **A. Quadir** and M. Tanveer. Robust Universum Twin Support Vector Machine for Imbalanced Data. *International Joint Conference on Neural Networks (IJCNN)*, Rome, Italy, 2025. doi:10.1109/IJCNN64981.2025.11228582.
- M. Sajid, M. Mushir, **A. Quadir** and M. Tanveer. RVFL-X: A Novel Randomized Network Based on Complex Transformed Real-Valued Tabular Datasets. *International Joint Conference on Neural Networks (IJCNN)*, Rome, Italy, 2025. doi:10.1109/IJCNN64981.2025.11227691.
- **A. Quadir**, M. Sajid, and M. Tanveer. One class restricted kernel machines. *31st International Conference on Neural Information Processing (ICONIP)*, Auckland, New Zealand, 2024. https://link.springer.com/chapter/10.1007/978-981-96-6954-7_17
- M. Sajid, **A. Quadir**, and M. Tanveer. Wave-RVFL: A randomized neural network based on wave loss function. *31st International Conference on Neural Information Processing (ICONIP)*, Auckland, New Zealand, 2024. https://link.springer.com/chapter/10.1007/978-981-96-6954-7_17

Awards and Honors

- IJCNN 2025 Doctoral Consortium, International Neural Network Society (INNS) 2025
- Gold Medal - B.Sc. (Hons.) Mathematics, Doon University 2020
- GATE, Mathematics (MA) - Gate Score: **548** 2022
- IIT JAM, Mathematics - All India Rank: **223** 2020
- IIT JAM, Mathematical Statistics - All India Rank: **246** 2020
- Gold Medal - B.Sc. (Hons.) Mathematics, Doon University, Dehradun 2020
- Qualified National Level B.Sc. Entrance Exam, Doon University, Dehradun 2017
- Merit-cum-Means Scholarship, Government of India 2015–2022

Academic Service and Leadership

- Organizing Committee Member - IEEE CIS Workshop on Cyber-Physical Systems: Research Outlook and Collaboration Opportunities, IIT Indore Feb. 27, 2026
- Organizing Committee Member - IEEE CIS High School Outreach Program, IIT Indore Sep. 27, 2025
- Organizing Committee Member - IEEE CIS Summer School on Emerging Trends in Computational Intelligence, Deep Learning and LLMs, IIT Indore Jul. 23–25, 2025
- Vice-Convener, CIS Membership Activity - IEEE CIS Chapter, MP Section Apr. 2025 – Present
- Organizing Committee Member - IEEE CIS Workshop on Innovation and Leadership in AI, IIT Indore Oct. 16, 2024
- Organizing Committee Member - IEEE Day Workshop on Membership Benefits of IEEE and IEEE Societies, IIT Indore Oct. 11, 2024
- Webmaster - IEEE CIS Chapter, MP Section Apr. 2023 – Apr. 2025
- Organizing Committee Member - IEEE CIS Workshop on Women in AI, IIT Indore Jan. 22, 2024
- Volunteer - EEG Training Workshop, IIT Indore May 05, 2023
- Volunteer - IEEE CIS Summer School on Deep Learning and Computational Intelligence, IIT Indore Dec. 12–16, 2022
- Volunteer - 29th International Conference on Neural Information Processing (ICONIP), India Nov. 22–26, 2022

Teaching Experience

- Teaching Assistant, MA207: Differential Equations II [Autumn 2025], Dept. of Mathematics, IIT Indore
- Teaching Assistant, MA205 & MA203: Complex Analysis [Autumn 2025], Dept. of Mathematics, IIT Indore
- Teaching Assistant, CS637: Theory of Computation and Graph Theory [Spring 2025], Dept. of CSE, IIT Indore
- Teaching Assistant, MA103N/105: Calculus-II / Calculus [Autumn 2024], Dept. of Mathematics, IIT Indore
- Teaching Assistant, MA217: Linear Programming [Autumn 2024], Dept. of Mathematics, IIT Indore
- Teaching Assistant, CS151: Computer Programming [Autumn 2023], Dept. of CSE, IIT Indore
- Teaching Assistant, MA204: Numerical Methods [Spring 2023], Dept. of Mathematics, IIT Indore

Workshops and Invited Presentations

- IEEE CIS Workshop on Cyber-Physical Systems: Research Outlook and Collaboration Opportunities, IIT Indore Feb. 27, 2026
- IEEE CIS High School Outreach Program on Foundations of Machine Learning and Artificial Intelligence, IIT Indore Sep. 27, 2025
- IEEE CIS Winter School on Computational Intelligence and Generative AI, IIT Indore Jan. 8-9, 2025
- International Joint Conference on Neural Networks (IJCNN 2025) - Rome, Italy Jun. 30 - Jul. 5, 2025
  (Three papers accepted and presented at this conference.)
- IEEE CIS Workshop on Innovation and Leadership in Artificial Intelligence, IIT Indore Oct. 16, 2024
- IEEE Day Workshop on Membership Benefits of IEEE and IEEE Societies, IIT Indore Oct. 11, 2024
- IEEE Day Workshop on Membership Benefits of IEEE and IEEE Societies, IIT Indore Oct. 11, 2024
- Symposium on Data for Public Good, 2nd Edition - IISc Bangalore Sep. 19-20, 2024
  (Selected for Talk in Young Researchers' Forum)
- IEEE CIS Workshop on Women in Artificial Intelligence, IIT Indore Jan. 22, 2024
- International Symposium on Emerging Trends in Deep Learning and Data Analytics (ET-DLDA) - South Asian University, New Delhi Jun. 15, 2023
- Second International Workshop on Core Data Science (CDS) - IIIT Allahabad May 23-27, 2023
- EEG Training Workshop - OPTIMAL Research Lab and IITI Drishti CPS Foundation, IIT Indore May 05, 2023
- IEEE CIS Summer School on Deep Learning and Computational Intelligence - IIT Indore Dec. 12–16, 2022
- Research Skills Development Workshop by Retd. Prof. N. Sundararajan, NTU Singapore - IIT Indore Sep. 08–10, 2022
- First International Workshop on Core Data Science (CDS) - IIIT Allahabad Sep. Aug. 29 - Sep. 02, 2022

Reviewing Activities

Journals: IEEE TNNLS; IEEE TFS; Pattern Recognition; Neural Networks; Neurocomputing.

Conferences: AAAI, NeurIPS; ICONIP; IJCNN; WCCI; ECCV.

Technical Skills

Programming & Scientific Computing: Python, MATLAB, C++, C, Java

Scientific Writing & Typesetting: LaTeX

Languages: English, Hindi, Urdu (strong reading, writing, and speaking competencies)